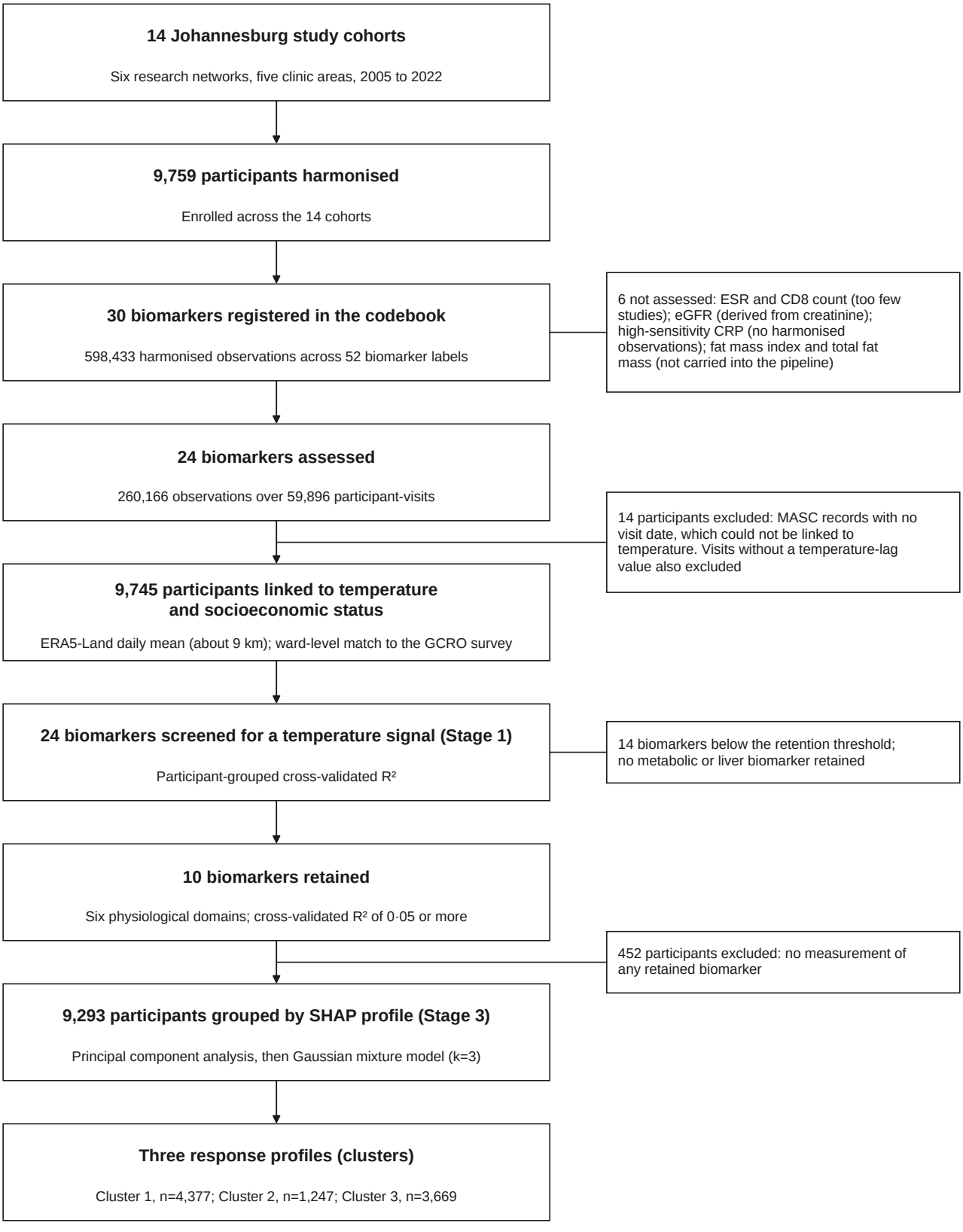
14 Johannesburg study cohorts
Six research networks, five clinic areas, 2005 to 2022
9,759 participants harmonised
Enrolled across the 14 cohorts
30 biomarkers registered in the codebook
598,433 harmonised observations across 52 biomarker labels
6 not assessed: ESR and CD8 count (too few studies); eGFR (derived from creatinine); high-sensitivity CRP (no harmonised observations); fat mass index and total fat mass (not carried into the pipeline)
24 biomarkers assessed
260,166 observations over 59,896 participant-visits
14 participants excluded: MASC records with no visit date, which could not be linked to temperature. Visits without a temperature-lag value also excluded
9,745 participants linked to temperature and socioeconomic status
ERA5-Land daily mean (about 9 km); ward-level match to the GCRO survey
24 biomarkers screened for a temperature signal (Stage 1)
Participant-grouped cross-validated R²
14 biomarkers below the retention threshold; no metabolic or liver biomarker retained
10 biomarkers retained
Six physiological domains; cross-validated R² of 0·05 or more
452 participants excluded: no measurement of any retained biomarker
9,293 participants grouped by SHAP profile (Stage 3)
Principal component analysis, then Gaussian mixture model (k=3)
Three response profiles (clusters)
Cluster 1, n=4,377; Cluster 2, n=1,247; Cluster 3, n=3,669